

DETAILED COURSE SYLLABUS

SOFTWARE TESTING

(AI-First Edition)

COURSE CODE: CSC13003 (CLC) / CS423

AI-FIRST EDITION - 2026

Applies OECD Working Paper 338 (Fuster Rabella, 2025) and the Post-AI Bloom Taxonomy (Hardman, 2025)

1. Course Overview

Item	Detail
Course title	Software Testing (AI-First Edition)
Course code	CSC15003 (CLC) / CS423
Credits	4 (45 lecture hours + 30 practical hours)
Course type	Major Selective - Software Engineering specialisation
Knowledge block	Major / specialisation
Year / semester	Year 3-4
Prerequisites	Object-Oriented Programming (OOP), Databases, Introduction to Software Engineering, Web Application Development
Teaching unit	Department of Software Engineering, Faculty of IT, HCMUS
Syllabus version	AI-first 2026 - aligned with the new teaching pack
Course SUT (System Under Test)	EShop - https://github.com/ttbhanh/eshop-sut (Node.js + React + React Native + SQLite)

2. Course Description

2.1 Short description

This course covers the foundations of software testing and AI-augmented testing: the testing process, test case design techniques (black-box, white-box, domain, state, decision table, use case), automation, performance, API/CI-CD, GUI/usability, security, DB testing, and mutation testing. The Project and most Homeworks are run against a single shared System Under Test - the EShop demo e-commerce application maintained by the Department. Students are required to use AI tools (of their own choice) for any task where AI is appropriate, and to audit each AI-generated artefact rigorously using the [AI-01]..[AI-06] template pack.

2.2 Position in the curriculum

This is a Major Selective course for the Software Engineering specialisation. It builds industry-ready Quality Assurance / Quality Control competence, with a particular focus on disciplined human-AI collaboration across the full testing life cycle - from test case design to CI/CD.

2.3 Update philosophy of the AI-first 2026 edition

Software Testing is one of the disciplines that industry has most aggressively shifted toward AI augmentation. The AI-first 2026 edition changes the teaching and assessment methodology while keeping the Department's traditional assessment weights intact:

- Total contact hours unchanged (45 lecture + 30 practical) and assessment weights unchanged (HWs 30% / Seminar 20% / Midterm 10% / Final 40%).
- New course learning outcomes G9.1..G9.6 added, drawn from the Post-AI Bloom Taxonomy (Hardman, 2025).
- All HW and Project work uses one shared System Under Test - EShop - so student submissions are directly comparable on the same scenario.
- The [AI-01]..[AI-06] template pack is mandatory with every AI-assisted submission.
- The Seminar is redesigned as 10 tool-focused tracks (T1..T10) following an 8-stage workflow (see section 5.4).
- The Department does NOT provide paid AI accounts. Students choose any tool (ChatGPT, Claude, Gemini, Copilot, Cursor - free tiers are fine).

3. Course Learning Outcomes

3.1 Traditional Bloom CLOs (unchanged)

On completion of this course, students will be able to:

Code	CLO description
G1	Explain the concepts and the testing process, and articulate testing's role in the software development life cycle.
G2	Apply test case design techniques (black-box, white-box, equivalence partitioning, BVA, decision table, state transition, use case).
G3	Produce a test plan, test cases, and bug reports to industry standards.
G4	Apply test automation with Selenium/Playwright/Cypress/JUnit.
G5	Conduct performance testing (JMeter/k6/Locust) and interpret results.
G6	Apply mock, API testing, CI/CD, and BDD in real projects.
G7	Use professional testing tools (Postman, JMeter, GitHub Actions, Cucumber).
G8	Present, justify, and defend a testing strategy before a panel or client.

3.2 New Post-AI Bloom CLOs (G9.1..G9.6)

Reference: Hardman, P. (2025) - Post-AI Learning Taxonomy. Applied to Software Testing:

Code	AI-Bloom Level	CLO description
G9.1	Understand	Ask AI to explain the ISTQB process; the student identifies at least 3 errors

		when cross-checking against the specification.
G9.2	Apply	Collaborate with AI to draft test cases for EShop; execute via Postman/Playwright; fix the cases AI missed.
G9.3	Analyse	Audit the AI test case set: V/I/IC verdict (Valid/Invalid/Incomplete); detect missing boundaries and duplicates.
G9.4	Collaborate	Team-level collaboration with an AI agent in a project: divide roles between AI and human; document who did what.
G9.5	Create	Build a new AI-powered plugin / script / harness that aids testing (e.g., assertion generator from mutation reports).
G9.6	Disrupt	Propose a Continuous AI Testing pipeline for an EShop microservice; identify trade-offs.

4. Lecture Content (45 hours / 11 weeks)

Each week is tagged with the appropriate AI-Bloom level and one of the 5 AI Use Categories: Cat.1 No-AI - Cat.2 AI for Preparation - Cat.3 AI Feedback - Cat.4 AI-Assisted - Cat.5 AI-Integrated. From HW02 onwards every homework runs on the EShop SUT.

Week	Topic	AI-first activity	CLO / Bloom	AI Cat.
1	Course intro, AI use categories, testing fundamentals	Demo AI generating test cases for a simple function; distribute [AI-01] AI Agreement.	G9.1 / 1. Understand	Cat.2/3
2	STLC, V-model, Black-/White-box overview	Workshop using Copilot/Cursor for pair-testing; students review AI-generated tests.	G9.2 / 2. Apply	Cat.4
3	Test case design - Equivalence, BVA, Decision Table (HW02 starts)	Required AI use for EShop FR-01..FR-04; students audit using [AI-02].	G9.3 / 3. Analyse	Cat.4
4	Domain Testing (paper workshop)	Cat.1 STRICT mode - students reason about domain partitioning unaided.	G2 / Bloom Apply	Cat.1
5	State Transition / Use Case Testing (HW03 - GUI Usability)	AI helps generate the EShop checkout state model; students find missing states.	G9.3 / 3. Analyse	Cat.4
6	Mid-Term Exam	Closed-book on paper - NO AI, NO materials.	-	Cat.1
7	GUI & Usability Testing - Seminar T1 (HW04 - Automation starts)	Seminar T1: Heuristic Showdown - Human vs AI (see /Seminar/).	G9.4 / 4. Collaborate	Cat.4/5
8	Web / Mobile / Desktop Automation - Seminars T2-T3-T4	Playwright/Appium + AI codegen + visual diff on EShop.	G9.2 / 2. Apply	Cat.4
9	Performance Testing - Seminar T5 (HW05 starts)	JMeter/k6; AI builds workload from HAR; students audit realism.	G9.3 / 3. Analyse	Cat.4
10	API + Contract + CI/CD + DB - Seminars T6-T7-T8 (HW06 starts)	AI tests from OpenAPI; AI flaky-test triage; AI data masking.	G9.4 / 4. Collaborate	Cat.4/5
11	Security + Mutation Testing + Review - Seminars T9-T10	ZAP/Semgrep + Stryker; AI assertion generation; course wrap-up.	G9.5/9.6 / Create/Disrupt	Cat.5

Final exam - closed-book on paper. Includes one 'critique an AI-generated test case' scenario drawn from the real EShop SUT.

5. Practical Content (30 hours)

Practical work has three strands - individual Homework (HW01..HW06) and team Seminar (PRJ-SM - 10 tracks T1..T10). All work integrates AI via the [AI-01]..[AI-06] templates.

5.1 Shared SUT - EShop

EShop is the demo e-commerce application maintained by the Department at <https://github.com/ttbhanh/eshop-sut>. Its four-tier architecture: Backend API (Node.js + Express + SQLite), Web Frontend (React + Vite + Tailwind), Web Admin, and Mobile App (React Native + Expo). The specification covers 22 Functional Requirements and 7 Security Requirements - full details in the repository README.

- Default test accounts: admin@eshop.com / Admin123! and test@eshop.com / Test1234!
- Local URLs: Backend localhost:3000 - Web localhost:5173 - Admin localhost:5174
- Students fork the repo to a personal account (HW01) or to the team repo issued by the instructor (from HW02 onwards).

5.2 Individual Homework - HW01..HW06

HW	Release week	Topic	Summary	Duration / CLO
HW01	Week 1	QA/QC Jobs / 20 Defects / Test 1 Physical Product	No SUT - students submit photos/video alongside their student ID card.	5h / G9.1, G9.3
HW02	Week 3	EShop / Domain Testing (Personal Seed)	Each student receives a unique FR pool based on (Student-ID mod N) to prevent copying.	10h / G9.2, G9.3
HW03	Week 5	EShop / GUI & Usability with real users	5-user remote test + Nielsen heuristics + AI vision review.	10h / G9.3, G9.4
HW04	Week 7	EShop / Automation (Hand vs AI)	Playwright + Mabl/Testim - maintainability comparison.	12h / G9.2, G9.3, G9.4
HW05	Week 9	EShop / Performance (AI Misinterpretation Hunt)	JMeter + k6; AI generates workload from logs; students audit AI conclusions.	12h / G9.2, G9.3, G9.4, G9.6
HW06	Week 11	EShop / API Testing (AI-spec-to-tests)	Postman + AI tests generated from OpenAPI; Pact contract test; AI critique.	12h / G9.2, G9.3, G9.4, G9.5

Every HW (except HW01) attaches the AI Template pack: [AI-02] Audit Report + [AI-03] Disclosure + [AI-05] Privacy Checklist. HW01 requires only [AI-03].

5.3 Team Seminar - PRJ-SM (10 tracks T1..T10 / 8-stage workflow)

Each team of 3-4 students chooses one of 10 tracks and follows the 8-stage workflow (see Seminar/Seminar_Guide.docx for full detail):

Stage	Stage name	Owner	Primary output
S1	Tool Survey & Proposal	Team (3-4 SV)	Proposal <= 1 page: 3 candidate tools,

			comparison, selection + AI Disclosure.
S2	Instructor / TA Review	Instructor / TA	Verdict within 2 working days: APPROVED / MINOR-CHANGE / RECLAIM.
S3	Deep Study + EShop Practice	Team	Install, work through tutorials, run at least one end-to-end scenario on EShop.
S4	User-Guide + Demo Screencast	Team	User_Guide.md (≥ 6 sections) + 5-8 minute screencast (≤ 100 MB).
S5	Pre-share (≥ 3 working days)	Team	Push materials to Moodle for the audience teams to review before the live session.
S6	Live Seminar (45 minutes)	Team + class	Pitch 10 min + Demo 10 min + Audience activity 20 min + Q&A 5 min.
S7	Audience Feedback	Audience teams	Minute-paper - one page per team (one thing learned / one thing unclear / 1-5 rating).
S8	AI Audit + Reflection	Team	Within 5 working days: [AI-02] + [AI-03] + [AI-04].

10 seminar topics (T1..T10) - each pairs a traditional tool with an AI direction:

Code	Topic	Traditional tools	AI direction
T1	GUI & Usability Testing	Maze / Hotjar / BrowserStack	AI heuristic review
T2	Web Automation Testing	Playwright / Selenium / Cypress	Mabl/Testim auto-heal / Copilot
T3	Mobile Automation Testing	Appium / Espresso / Detox	Applitools visual AI
T4	Desktop Automation Testing	WinAppDriver / Pywinauto / TestComplete	Image-based AI (Eggplant)
T5	Performance Testing	JMeter / k6 / Locust	AI workload synthesis
T6	API & Contract Testing	Postman / RestAssured / Karate	AI from OpenAPI / Pact AI
T7	CI/CD & Test-Harness Engineering	GitHub Actions / Jenkins / GitLab CI	AI flaky-test triage
T8	Database Testing	DbUnit / Liquibase / tSQLt	AI data masking / schema drift
T9	Security Testing (DAST/SAST)	OWASP ZAP / Burp / Semgrep	AI vulnerability triage
T10	Mutation Testing & Test Effectiveness	PIT / Stryker / mutmut	AI assertion gen / equivalent mutant

Each team also submits Seminar Slides (≤ 15 slides) + an Activity_Worksheet.md so audience teams can complete a 25-minute hands-on activity in class. Activity slide templates are in Seminar/InClass-Activities/.

6. Assessment

Component weights are unchanged - only the marking criteria and required attachments are updated.

6.1 Component weights

Code	Form	Weight	Marking notes (AI-first)	CLO link
HW 01-06	Individual	30%	Each HW carries a specific AI Use Category + required [AI-02]/[AI-03]/[AI-05].	G2, G3, G6, G9.1, G9.2, G9.3
Seminar PRJ-SM	Team	20%	One of the 10 tracks T1..T10 / 8-stage workflow / in-class activity.	G7, G8, G9.4, G9.5

Midterm	Individual	10%	Closed-book on paper / NO AI / Week 6.	G1, G2, G3, G9.3
Final	Individual	40%	Closed-book on paper / NO AI / end of term.	G1..G8 / G9.3

6.2 Required AI Template pack - [AI-01]..[AI-06]

Code	Template name	Description
[AI-01]	AI Agreement	Week 1 - defines the 5 AI Use Categories and the course AI policy.
[AI-02]	AI Audit Report	5 sections per artefact: prompt, AI output, V/I/IC verdict, reasoning, revised version.
[AI-03]	AI Disclosure Form	Submitted with every HW / Project - signed and attached. Missing = violation.
[AI-04]	Reflective Statement	Mandatory for Project / Seminar / Capstone - 300 words in English.
[AI-05]	Privacy Checklist	Self-check before submission - do not paste sensitive data into AI tools.
[AI-06]	Student Acknowledgement	Signed in Week 1 before any HW can be graded.

All six templates exist in both English (_En) and Vietnamese (_Vn) versions in /Homeworks/AI Templates/.

6.3 The 5 AI Use Categories

Cat.	Name	Description & where it applies
Cat.1	No AI	Midterm / Final / Week 4 Domain Testing paper workshop. STRICTLY FORBIDDEN.
Cat.2	AI for Preparation	Brainstorm and self-study only - AI does not produce the final artefact.
Cat.3	AI for Feedback / Revision	Critique or polish AFTER a complete draft exists.
Cat.4	AI-Assisted Production	AI generates parts; the student audits and integrates. Most HW work.
Cat.5	AI-Integrated Work	AI is a core methodology. Applied in Project and Seminar.

7. References

7.1 Primary references (unchanged)

- ISTQB Foundation Level Syllabus (latest version) - PRIMARY reference for the theory portion.
- Pressman, R. & Maxim, B. (2014). Software Engineering: A Practitioner's Approach (8th ed.). McGraw-Hill - Testing chapter.
- Myers, G., Sandler, C. & Badgett, T. (2011). The Art of Software Testing (3rd ed.). Wiley.
- Ammann, P. & Offutt, J. (2016). Introduction to Software Testing (2nd ed.). Cambridge University Press.
- Kaner, C., Bach, J. & Pettichord, B. (2001). Lessons Learned in Software Testing. Wiley.

7.2 New references on AI and Software Testing (2025+)

- Hardman, P. (2025). A Post-AI Learning Taxonomy. drphilippahardman.substack.com.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338. DOI: 10.1787/647880aa-en.
- Jia, Y. & Harman, M. (2011). An Analysis and Survey of Mutation Testing. IEEE TSE.
- Microsoft Research (2024). LLM-Assisted Test Generation - published case studies.

- Stryker / PIT / Applitools / Mabl - official documentation portals.
- DeepEval & Promptfoo documentation - testing frameworks for LLM systems (bonus reading).

7.3 Internal course materials (2026 edition)

- Weekly lecture decks W01..W11 (English) - /Slides_AI_First_2026/
- Homework pack HW01..HW06 (En + Vn) - /Homeworks/
- AI Templates [AI-01]..[AI-06] (En + Vn) - /Homeworks/AI Templates/
- Seminar pack - Guide + 10 topic briefs + 10 in-class decks - /Seminar/
- EShop SUT - <https://github.com/ttbhanh/eshop-sut> (README + api_specification.md + setup_guide.md)

8. Tools Used in This Course

8.1 Traditional testing tools

Tool category	Tools	Where applied
Test design / management	TestRail / Xray / Excel	Test plan and test case specifications.
Test automation (Web)	Playwright / Selenium / Cypress	Required for HW04 and Project automation.
Test automation (Mobile)	Appium / Espresso / Detox	Seminar T3 - Mobile Automation.
Performance testing	JMeter / k6 / Locust	Required for HW05 and Seminar
API testing	Postman / RestAssured / Karate	Required for HW06 and Seminar
CI/CD	GitHub Actions / Jenkins / GitLab CI	Seminar
Database testing	DbUnit / Liquibase / tSQLt	Seminar
Security testing	OWASP ZAP / Burp Suite / Semgrep	Seminar
Mutation testing	PIT / Stryker / mutmut	Seminar

8.2 AI-augmented tools - THE DEPARTMENT DOES NOT PROVIDE PAID ACCOUNTS

The Faculty of IT does NOT provide paid AI accounts to enrolled students. Students select their own AI tool from free tiers / trials (ChatGPT, Claude, Gemini, Copilot, Cursor, etc.) and are responsible for their own personal privacy, governed by [AI-05] Privacy Checklist.

Tool category	Tools (2024-2026)	Where applied
AI conversational	ChatGPT / Claude / Gemini	Brainstorm, evaluate test cases, audit AI output (Cat.3 / Cat.4).
AI in IDE	GitHub Copilot / Cursor / Windsurf	HW04 (Automation), Project automation (Cat.4).
AI testing platform	Mabl / Testim AI / Applitools	Seminars T1 / T2 / T3 - free tier or trial.
AI test generation	Diffblue Cover / Postbot	Seminars T6 / T10 - test and assertion generation.
AI flaky-test / triage	Datadog Flaky Test / Sealights AI	Seminar T7 - discuss only (paywalled).
AI for LLM / data	DeepEval / Promptfoo / Tonic.ai	Reference only - optional usage in PRJ-SM.

9. Course Regulations

9.1 General regulations

- Students missing more than 3 lectures or 1 practical session may not sit the final exam.
- HW / Project / Seminar marking follows the rubrics published at the start of term (see /Homeworks/ and /Seminar/).
- Academic dishonesty results in 0 marks for the relevant component or the entire course.
- Deadlines are strict - late submissions are not accepted (only critical justified cases with proof).
- Submission filename format: StudentID_ExerciseID_SelfAssessedGrade.zip (grade 000-100).
- All submissions must be TEXT-BASED (Markdown encouraged; LaTeX accepted; include a Save-As-PDF copy).
- At most 20 files per submission, each $\leq$ 20 MB. Use split-zip if necessary.

9.2 Regulations on AI use (updated 2026)

This section defines what counts as legitimate AI use and what counts as cheating:

9.2.1 Mandatory disclosure

- Every HW (except HW01) and every Project submission must attach [AI-02] AI Audit Report.
- Every HW must attach a signed [AI-03] AI Disclosure Form.
- Every Project, Seminar, and Capstone must attach an [AI-04] Reflective Statement.
- Failure to disclose = academic dishonesty.
- Misrepresenting the level of AI use (claiming less than reality) is equivalent to copying.

9.2.2 Allowed AI use by assessment type

Assessment	AI policy	Required attachments
HW 01-06	Allowed - disclosure required	Cat.4 by default - attach [AI-02] + [AI-03] + [AI-05].
Seminar PRJ-SM	AI required	Cat.5 - pair a traditional tool with an AI tool in the demo - [AI-02] / [AI-03] / [AI-04].
Midterm	STRICTLY FORBIDDEN	Cat.1 - closed-book on paper.
Final	STRICTLY FORBIDDEN	Cat.1 - closed-book on paper.

9.2.3 Department support

- The Department does NOT provide paid AI accounts - students choose their own free-tier tool.
- An 'AI for Testing' orientation session is delivered in Week 1 - attendance is mandatory.
- The instructor maintains the course materials repository (slide pack W01..W11, HW01..HW06, Seminar pack, AI Templates) - refreshed each term.
- The Moodle Discussion Forum is the channel for sharing good/bad prompts and AI tooling Q&A.

9.2.4 Anti-cheat - non-AI-replicable evidence

Each HW requires physical evidence that an AI cannot fabricate:

- HW01: photograph of the physical product and the student-ID card together in one frame.
- HW02-HW06: video of tests running on EShop plus a selfie with the debug screen visible.
- Project + Seminar: personal Git commit history (10-30% of students invited to a 5-7 minute oral defence at random).

10. Contact Information

Questions are best raised on the Moodle Discussion Forum or by emailing the teaching team:

- Dr. Lam Quang Vu - lqvu@fit.hcmus.edu.vn (Lecturer in charge)
- M.Sc. Tran Thi Bich Hanh - ttbhanh@fit.hcmus.edu.vn (Lecturer in charge)
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Syllabus for CSC15003 (Software Testing) / AI-first 2026 edition / Faculty of IT, University of Science, VNU-HCM / Last updated 2026-05.